SHIBAM BANIK

+91 9612706843 shibambanik0@gmail.com linkedin.com/in/shibam-banik

PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate specializing in **Blockchain Technology** with experience across **Software Development, AI/ML, and Cloud Technologies**. Proficient in **Python, C++, Java, React, Django, Node.js, and REST APIs**, with hands-on experience developing full-stack applications, **ML-based adversarial attack detection**, recommendation systems, and constraint-based simulation platforms. Strong foundation in **Data Structures & Algorithms, OOP, DBMS, Computer Networks, and Operating Systems**, with a proven ability to design, develop, and deploy practical software solutions. Seeking a **Software Engineer** role to contribute to scalable and innovative technology products.

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow, scikit-learn, Deep Learning, Computer Vision, NLP, Transformers, BERT, LSTM, Adversarial Robustness
Web: React.js, Node.js, Express.js, Django, REST APIs, FastAPI, HTML5, CSS3, JavaScript, Leaflet
Languages: Python, C++, Java, JavaScript, SQL **Databases:** MongoDB, PostgreSQL, SQLite
Cloud/Tools: AWS, Docker, Git, GitHub, Postman, VS Code, Jupyter Notebook, Vercel, Render

EXPERIENCE

National Institute of Technology (NIT), Sikkim – Research Intern (AI/ML) (Certificate) **Jun 2025 – Jul 2025**

Deep Learning, Adversarial Robustness, Computer Vision

- Developed a **PyTorch-based GhostNet pipeline** for multi-class adversarial attack detection on the CIFAR-100 dataset, achieving **80.5% classification accuracy**.
- Implemented **Local Histogram Equalization** and **Butterworth high-pass filtering** to improve robustness against adversarial perturbations, with comparative evaluation and hyperparameter optimization using **Optuna**.

PROJECTS

CineMatch – Movie Recommendation System (GitHub)

Flask, React.js, CountVectorizer, Cosine Similarity, TMDB API

- Developed a **content-based movie recommendation engine** using **CountVectorizer** and **cosine similarity** on the TMDB 5000 dataset.
- Built a **Flask REST API** with **auto-retrain-on-startup** fallback to handle large pickle files (∼185MB) on constrained hosting environments.
- Designed a cinematic **dark-themed frontend** and deployed on **Railway and Hugging Face Spaces** with **Python 3.11** pinning to resolve version conflicts.

LexSentinel – Legal Document Sentiment Analyzer (GitHub)

FastAPI, React.js, SQLite, HuggingFace Transformers, JWT, Render, Vercel

- Built a full-stack **legal document sentiment analyzer** using **FLAN-T5** to perform sentence-level classification into Positive, Negative, and Neutral categories.
- Developed a **FastAPI REST backend** with **JWT authentication**, user registration/login, and per-user analysis history, along with an asynchronous background job system for non-blocking document processing and real-time progress tracking.
- Built a responsive **React.js frontend** with drag-and-drop uploads, interactive **Recharts** visualizations, filterable sentence-level results, and CSV export; deployed the application using **Render and Vercel** with HuggingFace Inference API integration for memory-efficient inference.

EDUCATION

Vellore Institute of Technology, Vellore **2023 – 2027**
B.Tech in Computer Science and Engineering

Relevant Coursework: Data Structures, Artificial Intelligence, Cloud Computing, Database Systems, Computer Architecture, Cybersecurity, Data Privacy

CERTIFICATIONS

- **Gen AI Using IBM watsonx** – IBM Career Education, Issued June 2025